Yang Lab Report

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1. Model Selection Rationale

For object detection, we selected YOLOv5 and Faster R-CNN as our two architectures, representing single-stage and two-stage detection paradigms respectively.

YOLOv5 is a single-stage detector that predicts bounding boxes and class probabilities in a single forward pass over a grid-divided image. Its speed and strong out-of-the-box performance on small object detection make it well suited for acne lesion localization, where lesions are small relative to the full facial image. Redmon et al. demonstrated that single-stage detectors can achieve competitive accuracy at significantly faster inference speeds than two-stage alternatives [1].

Faster R-CNN, introduced by Ren et al. (2015), is a two-stage detector that first proposes candidate regions via a Region Proposal Network (RPN), then classifies each region independently [2]. While slower than single-stage methods, two-stage detectors typically achieve higher precision and are considered the standard academic baseline for object detection benchmarks. We use the ResNet-50 FPN backbone pretrained on COCO, replacing the classification head for our four acne classes plus background.

For classification in Part 2, we selected ResNet-50 as our backbone. He et al. (2016) demonstrated that residual connections allow training of much deeper networks without degradation, and ResNet-50 pretrained on ImageNet provides strong general visual features that transfer well to medical and dermatological imaging tasks [3].

References:

- [1] Redmon & Farhadi, YOLOv4 / YOLO series: https://arxiv.org/abs/2004.10934
- [2] Ren et al., Faster R-CNN (2015): https://arxiv.org/abs/1506.01497
- [3] He et al., Deep Residual Learning (2016): https://arxiv.org/abs/1512.03385

2. Preprocessing Steps

Part 1 — Detection

ACNE04 was downloaded in two formats from Roboflow: YOLO format for YOLOv5 training and COCO JSON format for Faster R-CNN. The dataset contains several thousand facial images across four acne lesion classes — nodules and cysts, papules, pustules, and whiteheads and blackheads — with bounding box annotations provided. No manual annotation was performed. The dataset was pre-split into train, validation, and test sets by Roboflow. Images were resized to 640×640 pixels for YOLOv5. Faster R-CNN used the original image dimensions with bounding box coordinates converted from COCO *xywh* format to *xyxy* pixel format during data loading.

Part 2 — Classification

A binary classification dataset was constructed from ACNE04 by extracting image patches using bounding box annotations. Positive patches were cropped from bounding box regions with 10 pixels of padding added on each side to include surrounding skin context. Negative patches were sampled randomly from image regions with IoU less than 0.1 with any ground truth box, ensuring they contained only clear skin. All patches were resized to 224×224 pixels to match ResNet-50's expected input size. A weighted random sampler was used during training to address class imbalance between positive and negative patches.

For domain adaptation, Reinhard color normalization was applied to training images by computing per-channel mean and standard deviation statistics in LAB color space over a reference set of DermNet images, then shifting ACNE04 patch statistics to match. Additionally, aggressive data augmentation was applied during training including random horizontal and vertical flips, rotation up to 20 degrees, color jitter across brightness, contrast, saturation and hue, Gaussian blur, and random erasing. These techniques aim to make the classifier robust to the lighting, color profile, and camera differences between ACNE04 and DermNet.

3. Training Details and Evaluation

YOLOv5

YOLOv5s was fine-tuned from COCO pretrained weights (yolov5su.pt) for 50 epochs with early stopping patience of 10 epochs, with the best checkpoint observed at epoch 17. Input images were resized to 640×640 pixels with a batch size of 16. The Ultralytics training pipeline was used with default SGD optimizer settings. The model with best validation mAP was saved and evaluated on the held-out test set.

Faster R-CNN

Faster R-CNN with a ResNet-50 FPN backbone was fine-tuned from COCO pretrained weights. The box predictor head was replaced to output predictions for 5 classes (4 acne types plus background). Training used SGD with learning rate 0.005, momentum 0.9, and weight decay 0.0005 for 20 epochs with a StepLR scheduler decaying by 0.5 every 5 epochs. Batch size was set to 4 due to the memory demands of the two-stage architecture. The model showed signs of overfitting early, with best validation loss observed at epoch 3–4, after which validation loss increased while training loss continued to decrease.

Classifier

ResNet-50 was fine-tuned in two stages. In Stage 1, the backbone was frozen and only the classification head was trained for 10 epochs at learning rate 5×10^{-4}. In Stage 2, all layers were unfrozen and trained end-to-end at learning rate 1×10^{-4} with early stopping. Cross-entropy loss with label smoothing of 0.1 was used throughout. The classifier was evaluated on the full DermNet test set using accuracy, F1-score, and AUROC.

Metric	YOLOv5s	Faster R-CNN
mAP@0.5	0.115	0.111
mAP@0.5:0.95	0.033	0.025
Precision	0.188	0.313
Recall	0.225	0.389

Metric	Without Adaptation	With Reinhard Norm
Accuracy	0.9220	0.9220
F1-Score	0.0000	0.0000
AUROC	0.5298	0.5403

4. Summary of Findings and Challenges

Findings

Comparing the two detection models, the results were notably close, with YOLOv5s achieving a marginally higher mAP@0.5 of 0.115 versus 0.111 for Faster R-CNN. However, Faster R-CNN demonstrated substantially higher precision (0.313 vs 0.188) and recall (0.389 vs 0.225), suggesting it localized more lesions correctly when predictions were made. This aligns with the expected tradeoff between single-stage and two-stage detectors — YOLOv5's speed advantage comes at a modest cost in localization quality on this dataset, while Faster R-CNN's two-stage design yields better per-box accuracy despite showing signs of overfitting due to the limited dataset size.

Cross-domain classification on DermNet revealed a severe domain gap. The classifier achieved 92.2% accuracy on the DermNet test set, however this reflects the class imbalance in DermNet (92% non-acne images) rather than genuine discriminative ability — F1-score was 0.000, indicating the model predicted non-acne for all test samples. AUROC of 0.540 with Reinhard normalization versus 0.530 without suggests the model retains marginal discriminative signal but cannot translate it into correct predictions at any threshold. Grad-CAM visualizations on DermNet samples showed diffuse, inconsistent attention patterns rather than focused activation on lesion regions, consistent with a model that has not successfully transferred its learned features to the new domain.

Challenges

Several challenges arose throughout the project. First, acne lesions are small relative to the full facial image, making precise localization difficult and contributing to lower mAP scores than seen on standard detection benchmarks. Second, significant class imbalance existed within ACNE04 — nodules and cysts were the most represented class with over 19,000 extracted patches, while whiteheads and blackheads had fewer than 2,400 — requiring weighted sampling to address during classifier training. Third and most significantly, the domain gap between ACNE04 consumer facial photographs and DermNet clinical dermatology images proved too large to bridge with color normalization alone. ACNE04 images are close-up, well-lit facial photos while DermNet images vary widely in zoom, lighting, and clinical context. Reinhard color normalization produced a marginal improvement in AUROC from 0.530 to 0.540, but did not meaningfully close the gap. More aggressive adaptation techniques — such as few-shot fine-tuning on DermNet samples, style transfer, or domain adversarial training — would likely be necessary for meaningful cross-domain performance.